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AN UPDATE ON ECONOMETRIC STUDIES OF ENERGY DEMAND BEHAVIOR

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INTRODUCTION AND SUMMARY

A by-product of the energy crisis has been a strong interest in modeling energy demand. Although the crisis has subsided in the last few years, the number of new demand studies has not. At least forty new econometric studies of energy demand have been published since the appearance of three recent and comprehensive surveys of the literature (1–3). It is useful to keep abreast of developments in the literature, as noted by Hartman (2), because the stock of information about demand continues to develop much like any other capital stock, with existing information in varying stages of obsolescence as new techniques and data bases are developed and exploited. Two kinds of information are conveyed in this body of literature: (*a*) estimates of how demand behaves, and (*b*) alternative ways of making the estimates. The information is sometimes bewildering, however, because of the variety of techniques employed and the disparities among the results.

Our primary interest in this survey, like that in Bohi (3), is the sensitivity of statistical results to differences in modeling techniques and sample data. This exercise can serve as a basis for evaluating and choosing among alternative procedures and alternative statistical results. The literature on this subject is inherently technical, making it difficult to avoid specialized terminology and concepts.[1] At the same time, nonspecialists are frequently the users of the information contained in econometric demand studies and should be aware of some of the technical issues that affect the reliability of that information.

[1] References are provided below for elaboration of the more technical issues.

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Only a subset of demand studies is examined here. In general, the range is restricted to partial equilibrium models of demand for individual energy products (i.e. electricity, natural gas, petroleum products, and coal) that are consumed by distinct sectors of the economy (i.e. residential, commercial, industrial, and transportation). Excluded, in particular, are models of substitution for energy as an aggregate commodity in production and consumption.[2] The list is further narrowed to studies based on the economic concept of demand (e.g. those that include a price effect) and generally to studies of demand in the United States. The material includes published papers and widely circulated unpublished discussion papers.

Three characteristics of the more recent group of studies covered here serve to distinguish it from the older body of literature, covered in Bohi (3). First, and perhaps most importantly, most of the newer studies use data taken from the period after 1974, when energy prices began to rise rapidly. Before 1974, energy prices were remarkably stable for extended periods of time, affording relatively few opportunities to observe adjustments in capital stocks and consumption rates in response to energy prices.[3] Many adjustments in consumption have occurred since 1974, some of which are well known—such as the efficiency of the automobile stock—and some of which are more subtle and less publicized.

Second, more-extensive data now exist on energy consumption in the residential sector at the household level. These data permit analysis of demand at a finer level of disaggregation—thus avoiding biases that result from averaging over broad groups of consumers—and of detailed characteristics of housing, appliances, and demographics that distinguish households. The principal source of the micro data is extensive surveys sponsored by the US Department of Energy, including the National Interim Energy Consumption Survey (NIECS) conducted in 1978–1979 (see 5) and the Washington Center for Metropolitan Studies (WCMS) surveys conducted in 1973 and 1975 for the Federal Energy Administration [see Newman & Day (6)]. In addition, the National Energy Conservation Policy Act of 1978 requires a building energy report prior to federal financial assistance, providing another source of detailed information about the residential housing stock [see Hirst et al (7)]. More detailed information at the state level on electric utility rates and consumption patterns comes from experiments conducted in response to the Public Utility Regulatory Policies Act of 1978. Many of the experiments focus on responses to

[2] For a review of this literature, see Kopp & Hazilla (4).

[3] Indeed, the once widely held view among electric utilities that price does not matter in load forecasting apparently confused the unimportance of price changes with the unimportance of price as a determinant of demand.

discriminatory pricing by time-of-day, in an effort to seek ways to reduce costs and conserve energy by shaving peaks in electricity demand. Studies of time-of-day demand characteristics are not included in this survey.[4]

Third, the studies covered here tend to be more sensitive to estimation problems inherent in the data and to the implications of alternative procedures for the results. Virtually all of the studies use a procedure other than ordinary least squares to adjust for violations of ideal conditions in regression models, and several compare the results of alternative estimators and functional forms. In short, the earlier emphasis on merely obtaining estimates of response coefficients has progressed to more careful analyses of the data.

Some characteristics of the newer group of studies remain unchanged, however. The emphasis is still on residential demand for electricity and (secondarily) natural gas, and on transportation demand for gasoline. There are still few studies of commercial and industrial energy demand, and those available continue to be severely hampered by the lack of detailed information on how energy is used in these sectors. There are also very few studies of coal demand, and of fossil fuels burned by electric utilities. As for modeling procedures, no new approaches to demand modeling serve to distinguish this group from earlier studies. Also like the older studies, the new studies show relatively little concern for separating the influence of supply from demand in observations on market data. Ignored altogether are supply considerations for energy-using capital stocks.

The following sections briefly describe the individual studies and statistical results, organized according to consuming sector and type of demand model. On the basis of this review, we present general comparisons of model results. Overall, we have found that the studies covered here use more sophisticated methods and better data than earlier empirical studies. However, our knowledge of demand behavior has not improved appreciably in the weakest areas. Commercial and industrial demand behavior remain largely in the dark, and additional analyses of the same aggregate level data are not likely to be illuminating. Estimates of price elasticities of demand for gasoline, residential electricity, and residential natural gas are more reliable and compare closely with the findings of earlier studies. The evidence suggests that the energy crisis has not caused an abrupt change in these elasticity parameters. Additional details about household consumption behavior, apart from price and income effects, have come to light because of the availability of more data. Howevcr, these details do not fall within the scope of this paper and readers are referred to the original works.

[4] See (8) for a survey of this subset of the literature.

MODELING ISSUES AND OPTIONS

It is useful to begin with a brief description of the more important issues in modeling energy demand and the approaches modelers taken in confronting these issues. The demand for energy is derived from the demand for the services energy provides, and, in almost all cases, a unit of capital is required in association with a unit of energy to produce the desired service. These two characteristics of energy demand give rise to a number of basic modeling options. Model structure, along with differences in the approach to other modeling issues, serve to distinguish the studies reviewed below and partially account for disparities in their results.

Model Structure

The consumption of any given fuel in conjunction with a specialized stock of capital equipment—household appliances, automobiles, etc—can be represented by the following identity:

$$Q_{\mathrm{i}} \equiv \sum_{k=1}^{M} R_{k\mathrm{i}} A_{k\mathrm{i}}. \qquad 1.$$

This equation states that total consumption of fuel i is the sum of fuel consumed by each type of equipment in the capital stock, A. $R_{k\mathrm{i}}$ can be interpreted as the utilization rate of the k^{th} type of equipment of the i^{th} fuel.

A structural model of energy demand behavior explicitly recognizes the derived nature of the demand by specifying separate demand functions for the equipment stock and the utilization rate.[5] Typically, the behavioral relationships are identified separately:

$$A_{\mathrm{i}} = f(P_{\mathrm{i}}, P_{\mathrm{j}}, P_{\mathrm{a}}, Y, X); \quad \mathrm{i} \neq \mathrm{j}. \qquad 2.$$

$$R_{\mathrm{i}} = g(P_{\mathrm{i}}, Y, Z). \qquad 3.$$

Equation 2 indicates that the demand for equipment using fuel i depends on the price of fuel i, the price of alternative fuel j, the price of the equipment P_{a}, income Y (or value added, in the case of a firm), and a vector of other variables X (such as household size). The utilization rate specified in Equation 3 depends on the own-price P_{i}, income Y, and a vector of other variables, Z. P_{j} is not a relevant argument since the fuel type is constrained by the appliance chosen (unless the equipment has dual fuel-burning capacity).

[5] As used in this paper, the term "structural" refers to a model in which the endogenous elements of demand are specified explicitly. This is analogous to the more frequent use of the term applied to models that specify separate supply and demand relationships for the market. For our purposes, models that combine the endogenous elements of demand in a single reduced form equation are referred to as reduced form models, regardless of whether or not the supply side of the market or the market price is endogenous.

The interdependence of these two functions illustrates the joint nature of the decision process and indicates that both the capital stock and the utilization rate are endogenous factors of energy demand. We refer to models that simultaneously estimate the level of capital stock and the level of energy consumption as "structural."

If information on equipment stocks is unavailable, the model represented by Equations 1, 2, and 3 must be simplified, with an inevitable sacrifice of model detail. When stock data are available, but not data on stock prices, a reduced-form end-use model is an option. In an end-use model the equipment stock (or saturation rate) is held constant at the observed level without any attempt to explain what determines that level, in order to focus on the determinants of the utilization rate, conditional on the equipment stock. The identity equation (Equation 1) is used to obtain a weighted sum of the contribution of each unit of equipment for total fuel consumption. This approach is frequently referred to as the Fisher-Kaysen (9) model.[6]

In the absence of equipment stock data, another option is to collapse functions 2 and 3 into a single reduced-form equation, based on the generalization of Equation 1:

$$Q_{\mathrm{i}} = h(A_{\mathrm{i}}, R_{\mathrm{i}}) \qquad 1'.$$

where A_{i} and R_{i} are vectors of individual equipment types using fuel i. Substituting Equations 2 and 3 into Equation 1′ yields the reduced-form equation,

$$Q_{\mathrm{i}} = k(P_{\mathrm{i}}, P_{\mathrm{j}}, Y, X, Z); \qquad \mathrm{i} \neq \mathrm{j}. \qquad 4.$$

A variation of Equation 4 is sometimes used when equipment stock data are available:

$$Q_{\mathrm{i}} = b(P_{\mathrm{i}}, P_{\mathrm{j}}, Y, X, Z, A). \qquad 4'.$$

Equation 4′ differs from an end-use model in that the appliance stock is not tied to consumption by the identity equation (Equation 1) and end-use elasticities cannot be calculated.

Because Equations 4 and 4′ fail to capture the relationship between the utilization rate and the equipment stock, the distinction between short and long run adjustments is blurred. Hence, models of this type are referred to as reduced-form static models. Frequently, the distinction between short and long run adjustments is inferred from the type of data used, where time-series observations are used to measure short-run adjustments and cross-

[6] Fisher & Kaysen (9) developed this basic model in 1962. Both the structural and reduced form end-use models are also commonly referred to as "conditional demand" models since the short run results are conditional upon a given capital stock.

section data are used to measure long-run adjustments. However, the distinction achieved by this procedure is not very satisfactory (10).

To overcome this deficiency, another device is commonly used to capture the dynamics of the demand process. A distinction is made between actual consumption Q_t and desired consumption Q_t^* at time t, where the difference reflects the difficulty of adjusting equipment stocks to the desired configuration warranted by a change in relative fuel prices (or other variables). A partial or flow adjustment process is assumed

$$Q_t = Q_{t-1} + d(Q_t^* - Q_{t-1}) \qquad 5.$$

where d lies between zero and unity to reflect the partial adjustment in actual consumption toward desired consumption.

With Q_i in Equation 4 expressed in terms of desired consumption Q_t^* of fuel i, the unobserved variable may be eliminated by substituting Equation 5 into Equation 4 to yield an expression in terms of the observable variable Q_t. The result is an equation that includes the lagged value of consumption Q_{t-1} as an explanatory variable. When Equation 4 is expressed in linear form, the coefficient of Q_{t-1} gives an estimate of the adjustment parameter, which may be made to distinguish short from long run response coefficients.[7]

Each of these models—structural, reduced-form end-use, reduced-form static, and reduced-form dynamic—have inherent advantages and weaknesses. The structural model incorporates the greatest amount of theoretical information about the nature of energy demand behavior and empirical detail. It is also the most cumbersome to estimate. End-use models retain the distinction between capital stock and the utilization rate and can provide estimates of separate end-use elasticities, but are limited to conclusions about the short run. The reduced-form static model provides the least amount of information, but it is relatively simple to estimate. The flow adjustment or reduced-form dynamic model expands on the static version in a potentially useful way, but the distinction between short and long run adjustments is generally based on an arbitrary specification of the adjustment process. The choice of approach is governed by the objective of the analysis, the availability of capital stock information, and the ease with which the model is employed.

Other Modeling Issues

In addition to their basic approaches to demand, models differ in level of aggregation, supply considerations, measurement issues, functional form, and estimation technique.

[7] See Berndt et al (11) for a discussion of this procedure and the implications for estimation. Note that alternative specifications of the adjustment prices yield lagged adjustment models of the same general form.

LEVEL OF AGGREGATION The level of aggregation in the data has been found to make a difference in the estimated results of demand models, with models estimated at finer levels of aggregation performing better than their more aggregated counterparts (3). This result is expected because more aggregated data average over disparate characteristics of economic units and reduce the amount of pertinent information in the sample. The disparities among customers are particularly pronounced in the industrial sector, where different plants in the same industry use different production processes and industry groupings combine widely different commodities and processes.

The opportunity for analysis at the level of the economic unit is largely limited to the residential sector, where survey data on household characteristics and energy consumption patterns are available. Here, too, the data are concerned primarily with electricity consumption and only secondarily with natural gas and petroleum products. Data on manufacturing energy consumption are restricted to states and regions, and to two-digit and three-digit SIC (standard industrial classification) manufacturing categories. No possibility currently exists to analyze consumption behavior at the firm or plant level. Data are available on fuels burned by electricity-generating plants, enabling more detailed analysis of this sector.

SUPPLY CONSIDERATIONS The influence of supply considerations in energy demand models, the second category that serves to distinguish among the studies, refers to the traditional identification problem in econometrics (12). The problem is essentially one of inferring characteristics of demand alone from market data that reflect the combined influences of both supply and demand behavior. The problem is nonexistent only if supply is perfectly elastic; otherwise, there is a relationship between quantity supplied and price that must be taken into account in estimates of the relationship between price and quantity demanded.

The traditional approach to identification is to specify a supply model along with a demand model, and to estimate the models jointly or separately with a suitable simultaneous equation technique on the assumption that the usual "exclusion restrictions" for identification are satisfied.[8] The exclusion restrictions usually refer to exogenous variables that serve to distinguish shifts in supply from shifts in demand. That is, variables that influence supply but not demand, and vice versa, are excluded variables that distinguish demand relationships from supply relationships in the data. The restrictions may not be appropriate in models of energy demand (or supply) because decisions are intrinsically dynamic

[8] The exclusion restrictions are established prior to estimation and hence are not testable. See Fisher (12) for a discussion of exclusion restrictions.

and depend on expectations about future prices (and other variables). As Sargent (13) demonstrates, it is difficult to specify excluded variables in supply and demand equations if decision-makers form expectations about future prices on the basis of all available information. Variables that influence supply will also influence demand, and vice versa, if the variables also influence expectations about future prices. Sargent demonstrates an alternative procedure for identification involving cross-equation restrictions among disturbance terms, but to date the procedure has not been used to estimate models of energy demand.[9]

The influence of supply enters models of energy demand in two dimensions. The most straightforward is the supply of the fuel in question, and a few models attempt to account for the interdependence between price and quantity in demand models. Less obvious, and yet to be accounted for, is the supply of energy-using capital equipment associated with energy demand. In effect, all studies implicitly assume that the supply of equipment is perfectly elastic.

MEASUREMENT ISSUES Measurement issues represent the third category where choices must be made in modeling energy demand. A number of issues are relatively mundane. For example, weather-related variables are normally summarized in measures of the number of heating and cooling degree days, though the measures are imperfect in capturing regional weather differences. Income of households (or heads of households) can be measured according to earnings or expenditures, although the latter is generally considered the better measure of permanent income. Other scale variables are intuitively important, such as the size of the home, number of rooms and types of appliances. Yet, these variables are also correlated with income and each other, making it difficult to infer their separate influences.

Similar problems occur with scale variables in the manufacturing sector, where the level of output is frequently measured by an index of value added for the sector. Aggregative measures of output reflect the combined influence of changes in the level of output and changes in the mix of output, and the relative importance of each cannot be separated. A change in the output mix may be price-induced because higher energy costs affect sales of relatively energy-intensive commodities, causing a substitution in consumption in addition to substitution among inputs in production.

One of the more vexing measurement problems concerns the choice among marginal, average, and inframarginal prices of electricity and natural gas. The marginal price of electricity is generally considered the most important in consumption decisions, and is more commonly included

[9] For application to energy-supply models, see Epple & Hansen (14), Epple (15), and Roberds (16).

in recent demand models than in earlier studies, yet the issue is far from resolved. Most residential consumers, it is argued, have no idea of marginal rates in block schedules, and are more likely to adjust consumption according to their perceptions of the average price. Furthermore, in studies using data that aggregate over individual consumers and rate schedules, the measure of the marginal price is difficult to define. In any case, the rate schedule contains more information than either the average or marginal price, including fixed charges and inframarginal rates, and the relative importance of this information is yet to be established.[10]

FUNCTIONAL FORM The choice of functional form is another decision facing demand modelers. Log-linear, linear, and translog are the most popular options. The log-linear form is convenient because parameter estimates measure elasticities directly, while linear forms are preferred by those who question the reasonableness of assuming constant elasticities at all price levels. Linear and log-linear forms are restrictive in their assumptions about underlying utility functions of households and production functions of firms. In particular, the underlying functions must be linear, implying that elasticities of substitution in consumption and production are constant and equal (17). The functional form, in addition, imposes restrictions on the data that can affect parameter estimates (18).

The translog is one of several so-called flexible functional forms developed to relax the restrictions imposed by linearity. While more flexible in form, the translog also has limitations.[11] Two limitations are particularly troublesome in energy demand models. First, and most important, the translog models are static and hence unable to capture the intertemporal properties of demand. Second, the procedure generates a large number of parameters so that, to preserve degrees of freedom, restrictive assumptions are required to reduce the number of independent parameters. Below, the results of studies using the translog form may be compared with the simpler linear forms. Some studies estimate more than one functional form for purposes of comparison.

ESTIMATION TECHNIQUES The final item in the list of procedures that distinguish among the studies is the estimation technique. A variety of estimators are employed, including ordinary least squares (OLS), generalized least squares (GLS), two-stage least squares (TSLS), seemingly unrelated regression (SUR), and error components (EC).[12] Some studies

[10] For an assessment of the relative merits of marginal and average price specifications, see Taylor (1) and the response by Nordin (22). See National Economic Research Associates (23) for a discussion on including the entire rate schedule.

[11] See Kuh (19) and Berndt et al (20) for discussions of the limitations of the translog.

[12] See Theil (21) for discussion of these estimators and for additional references.

apply a variety of estimators to compare results, and a few studies fail to report the estimator used.

Variations of the GLS procedure are the most frequently employed estimators because of widespread prior beliefs about possible correlations in the error terms. The EC method is now commonly used in studies with pooled data sets in order to account for typical cross-section and time-series correlation patterns. SUR is frequently used in models with separate equations for fuel types and end uses to account for correlations among error terms in the separate equations, while TSLS is popular in models with explicit specifications of simultaneous relationships. In general, the ideal conditions for OLS estimators are assumed to be violated in energy demand analysis, and OLS results are usually provided merely for purposes of comparison.

RESIDENTIAL ENERGY DEMAND

The residential sector has been studied more extensively and more diversely than the other sectors and thus provides the best opportunity to compare and contrast modeling results. The models are grouped first according to the fuel type under consideration and then by the modeling procedure employed.

Electricity Demand

REDUCED-FORM STATIC MODELS The seven separate studies listed in Table 1 use reduced-form static models of the type represented by Equation 4. The studies are further grouped according to the use of aggregated or disaggregated data and according to whether or not an average or marginal price of electricity is used. The table also indicates the sample period, estimation method, and list of explanatory variables in addition to price and income. All of the models use linear or log-linear functional forms.

Lyman (24) and Smith (25) both use aggregated data and average prices of electricity, but differ substantially according to the type of sample and the specification of the equation. Both employ pre-1973 data from selected utilities, but Smith runs separate time-series regressions for each utility surveyed, while Lyman uses pooled observations by geographic region. The range of estimated price elasticities are similar for each study, but vary widely by utility and region, from highly inelastic to highly elastic, regardless of the estimation technique chosen. Both authors suggest that structural differences in their sample sets account for the observed variation, but they do not test for the causes. Smith is interested in determining whether or not the use of average or marginal price alters the results significantly, and he concludes that no appreciable difference can be detected. Lyman is the only study reviewed that uses parametric variable transformations to determine the "appropriate" functional form. He concludes that the independent variables (but not the dependent variable) should be measured in log form in residential energy markets.

The next two studies also use aggregate data, but differ in their use of a marginal price measure. As with the first two studies, Houthakker's (26) estimates for separate regions using a pooled time series of state data yield erratic price elasticities, leading the author to conclude that the static results are less acceptable than are his dynamic results, presented below. Wills (27) obtains small[13] long-run price elasticities from a cross-section of "typical customers" from utility service areas in Massachusetts. He finds a significant difference in the price elasticities of all-electric and other households, with all-electric households having a smaller elasticity. This result suggests that interfuel substitution is important in homes with multiple-fuel hookups. Wills does not account for shifts of households between the two categories and his results are therefore somewhat less than long run.

The same type of model applied to household-level data produces less-elastic price and income responses and, except for Garbacz (28), substantially greater stability across different samples and models. Hirst et al (29) examine a cross-section of US households from the NIECS survey, and Iowa (30) uses a pooled sample of nonfarm Iowa households. The first study uses a measure of the average price of electricity, while the second uses marginal price. Hirst et al (29) report long-run estimates for all households, while Iowa (30) reports separate estimates for selected demographic groups based on age, income, and consumption patterns.

The price elasticities obtained by Garbacz (28) are markedly smaller than those of the other two studies, even though the sample of households is drawn from the same source as Hirst et al. The discrepancy appears to be explained by the inappropriate use of both average and marginal prices in the same model.[14]

From this group of studies one would conclude that the long-run price elasticity is about 0.6 and the long-run income elasticity is less than 0.2. The substantially greater importance of price relative to income in the demand for electricity is a bit deceiving because of the presence of income-related variables in most equations. The indirect effect of income through measures of housing characteristics and appliance saturation adds to the direct effect of income on consumption. These studies also yield a wealth of information

[13] References to the magnitude of price elasticities are in terms of absolute value.

[14] See Nordin (22) for a discussion of appropriate measures to use in combination with the marginal price.

Table 1 Residential electricity

Study (date)	Sample[a]	Price elasticity[b] Short	Price elasticity[b] Long	Income elasticity Short	Income elasticity Long	Regression procedure[c]	Other variables[d]
Reduced-form, static							
Agg., ave. price							
Lyman (1978)	1959–68; P; 10 regions (utility)	−0.13 to −1.36		−2.14 to 0.55		MLE	PNGas, D, W, YDIst
Smith (1980)	1957–72; T; 27 utilities (utility)	−0.07 to −1.56		0.59 to 1.76		OLS	W, PNGas
		−0.11 to −1.57		0.21 to 1.62		TSLS	
Agg., marg. price							
Houthakker (1979)	1964–76; P; US (state)	−1.18		1.39		WLS	W, IMC, APNGas
	1964–76; P; Northeast states	n.s.		1.67			
	1964–76; P; N. Central states	n.s.		1.45			
	1964–76; P; Southern states	−1.07		1.58			
	1964–76; P; Western states	−0.56		1.79			
Wills (1981)	1975; C; Massachusetts (Lypteal customer)	n.s.	−0.27	n.s.		WLS	IMC, PNGas, W, H (all customers)
			−0.18				(electric heating customers)
			−0.52				(others)
Disagg., ave. price							
Hirst et al (1982)	1978–79; C; US (household)		−0.71[e]		0.12[e]		H. W. D (linear model)
			−0.67		0.16		(log model)
Disagg., marg. price							
Iowa (1982)	1977–80; P; Iowa (non-farm household)	−0.55 to −0.71		0.13		—	W. D. AElec (monthly observations)

Garbacz (1983)	1978–79; C; US (household)		−0.05		0.14	—	APElec, D, G, PFuel
Reduced-form, dynamic							
Agg., ave. price							
Uri (1978)	1971–75; T; US (US)	−0.35	−0.35	2.00	2.04	SDR	W, E, No. of customers (monthly observations)
Beierlein et al (1981)	1967–77; P; Northeast (state)	−0.11	−1.87	n.s.	n.s.	OLS	APNGas, POIl
		−0.11	−2.20	n.s.	n.s.	EC	
		−0.09	−2.19	n.s.	n.s.	EC-SUR	
Maddigan et al (1983)	1969–78; P; N.E. states (rural state)	−0.20	−0.47	0.42	0.98	TSLS	POIl, W, no. of customers, farming activity, G
	S.E. states	−0.22	−0.50	0.32	0.70		
	N. central states	−0.21	−0.93	n.s.			
	S.W. states	−0.16	−0.74	n.s.			
	Western states	−0.13	−0.26	0.18	0.37		
	1969–78; P; N.E. states (total state)	−0.18	−0.62	0.19	0.66		PNGas, W, no. of customers
	S.E. states	n.s.		n.s.			
	N. central states	−0.25	−0.71	n.s.			
	S.W. states	−0.10	−0.26	0.30	0.76		
	Western states	n.s.		0.24	1.23		
Agg., marg. price							
Houthakker (1979)	1964–76; P; US (state)	−0.11	−1.42	0.14	1.78	WLS	W, Y-IMC, APNGas
	1964–76; P; N.E. states	−0.14	−0.67	0.62	3.00		
	1964–76; P; N. central states	−0.11	−2.50	n.s.	0.32		
	1964–76; P; Southern states	−0.08	−1.11	0.15	2.01		
	1964–76; P; Western states	−0.11	−1.12	0.15	1.50		

Table 1 (*continued*)

Study (date)	Sample[a]	Price elasticity[b] Short	Price elasticity[b] Long	Income elasticity Short	Income elasticity Long	Regression procedure[c]	Other variables[d]
Blattenberger et al (1983)	1960–75; P; US (state)	−0.10	−1.05	0.08	0.80	VC	IMC, PNGas, POil, W, gas supply
		−0.09	−1.21	0.06	0.74		IMC, PNGas, POil, W, gas supply, E
Reduced-form, end-use							
Agg., marg. price							
Hartman & Werth (1981)	1960–75; P; US (state)	n.s.		n.s.		RE, PE	IMC, AElec, no. of customers
Disagg., ave. price							
Parti & Parti (1980)	1975–76; C; San Diego County (household)	−0.58		0.15		TSLS	H, W, AElec, D (weighted ave., of monthly observations)
Disagg., marg. price							
Acton et al (1980)	1972–74; P; Los Angeles (meter book)	−0.70		0.40		WLS	PGas, IMC, AElec, ANGas, H, D, W (bi-monthly observations)
	1972–73; P; Los Angeles	−0.52		0.31			
	1973–74; P; Los Angeles	−0.20		0.42			
	1972–74; P; Los Angeles	−0.35		0.38			

Barnes et al (1981)	1972–73; C; 23 metro areas (household)	−0.55		0.20		IV	IMC, AE, W, H, G, D
		−0.88		0.21		OLS	
Archibald et al (1982)	1975; C; US (household)	−0.40		0.11		OLS	D, W, H, AElec, Y-IMC (weighted ave. of monthly observations)
		−0.47[f]					(Peak)
		−0.32					(Off-peak)
Structural							
Disagg., ave. price							
Garbacz (1983)	1978–79; C; US (household)	−0.19	−1.40	0.10	0.41	TSLS	PFeul, AElec, W, D, G
Disagg., marg. price							
Garbacz (1982)	1978–79; C; US (household)	−0.67	−1.51	0.13	0.36	TSLS	AElec, W, D, PFuel, G, IMC
Dubin & McFadden (1982)	1975; C; US (household)	−0.20		0.03		OLS	(all electric)
		0.02		0.08			(other)
		−0.16		0.06			(total)
		−0.31		0.01		IV	(all electric)
		0.04		0.02			(other)
		−0.20		0.01			(total)

[a] The term in parentheses refers to the unit of observation. T = time-series data; C = cross-sectional data; P = pooled data.

[b] n.s. = not significant at the 5% level.

[c] MLE = maximum likelihood estimation; OLS = ordinary least squares; GLS = generalized least squares; WLS = weighted least squares; EC = error components; VC = variable components; RE = random effects; FE = fixed effects; SUR = seemingly unrelated regressors; TSLS = two-stage least squares; IV = instrumental variables.

[d] PElec = price of electricity; APElec = average price of electricity; PNGas = price of natural gas; APNGas = average price of natural gas; POil = price of oil; PGas = price of gasoline; PCoal = price of coal; PFuel = price of local alternative fuel; IMC = inframarginal charge; AElec = stock of electricity-using appliances; AGas = stock of gasoline-using appliances; Auto = stock of automobiles; Truck = stock of trucks; CPI = consumer price index; YDist = income distribution; POP = population; D = demographic variables; H = housing stock and characteristics; W = weather variables; G = geographic variables (including regional dummies); E = embargo dummy; Employ = rate of (un)employment; Tax = import tax on equipment stock.

[e] The authors interpret long run estimates as lower bound values due to inclusion of housing stock variables.

[f] −0.47 and −0.27 are reported for peak and off-peak months respectively. The authors' averages are not weighted by monthly consumption, however.

on detailed characteristics of demand by season, demographics, and housing characteristic that is not possible to summarize here.

REDUCED-FORM DYNAMIC MODELS Five studies are represented in this group of dynamic models, all of which specify a demand equation with a lagged dependent variable generated from an adjustment process such as that described in Equation 5. In addition, all five studies use aggregate-level data, although they differ in the unit of measurement. Two of the studies specify marginal prices of electricity, and three specify average prices.

The short-run price and income results do not appear to differ according to whether average or marginal prices are used, as indicated by comparing the results of Beierlein et al (31) and Maddigan et al (32) with those of Houthakker (26) and Blattenberger et al (33). These four studies are quite similar in sample period, use of pooled state data, and specification of functional form and other independent variables. The short-run price coefficients are all quite small and the calculated long-run elasticities are quite large. Also, the results do not change much according to the estimation method employed.

The principle difference among the samples is that Maddigan et al (32) limit their initial sample to rural portions of each state (often under-represented in other studies). As might be expected from the similarity of their results with those from other studies, these authors find little evidence of a significant difference in rural and nonrural elasticities.

There is, however, considerable variation in the estimated long-run elasticities even though short-run elasticities are roughly similar. The variation results from differences in the estimated adjustment coefficients. Although the coefficients are qualitatively consistent (that is, most of the adjustment occurs in the long run), the quantitative differences are enough to cause erratic long-run elasticities. Consequently, the long-run results might be viewed with skepticism.

Uri's (34) reduced-form dynamic estimate of the short-run price elasticity, using monthly time-series data for 1975 and an average price specification, is considerably larger than that of the other studies. The adjustment coefficient is nearly zero, implying virtually instantaneous adjustment. However, since the period of observation is only one month, there is little basis for using a flow-adjustment model. The reported income elasticity is also exceptionally large.

REDUCED-FORM END-USE MODELS Each of the five studies reviewed in this group assumes that utilization rates vary across end uses, and that end-use elasticities should be estimated separately. Although the importance of the appliance stock is acknowledged, appliance demands are not estimated. In each model, aggregate elasticity estimates are weighted averages of the elasticities for each appliance type and are therefore conditional upon the composition of the appliance stock.

Hartman & Werth (35) employ the only aggregated data sample used and obtain statistically insignificant results. In contrast, the other four studies use household data and obtain significant and consistent price responses. These results tend to substantiate those from the earlicr group of reduced-form models estimated with household-level data.

The disaggregated studies incorporate a wide range of sample characteristics, estimators, and variable specification. In particular, Parti & Parti (36) use an instrumental variable estimation of average price, while the other authors—Acton et al (37), Barnes et al (38), and Archibald et al (39)—use a marginal price specification and the inframarginal charge variable suggested by Nordin (22) to separate income and own-price effects. Owing to the large amount of detailed information required, the samples are drawn from cross-sectional surveys. Archibald et al (39) employ a national survey [WCMS (see 6)], while the others limit their analyses to metropolitan areas. Importantly, the results do not appear to be sensitive to either the specification of price or to the transition from pre- to post-1973 electricity price levels.

The individual estimates by appliance are not as consistent as the aggregate estimates just described. They tend to vary widely among the studies, sometimes differ with engineering data,[15] and are frequently implausible or insignificant. Archibald et al and Parti & Parti calculate elasticities by month and find that they are higher during months of peak demand.[16]

STRUCTURAL MODELS Three studies are represented that employ variations of the structural model given by Equations 2 and 3, where the distinction between utilization rates and changes in capital stocks is explicit and the capital stock is endogenously determined. The two studies by Garbacz (40, 41) use a three-equation model in which electricity consumption, electricity price, and appliance stock are endogenous variables. The model is estimated with household-level data from the 1978–1979 NIECS surveys (see 5). Dubin & McFadden (42) use a cross-section data set from the 1975 WCMS household survey (see 6).

The two studies by Garbacz do not distinguish among end uses of electricity, but include an index of overall appliance stock size. Both papers use the same model and the same NIECS household sample, yet yield

[15] See Hartman & Werth (35) for a review of engineering appliance use estimates.

[16] For Parti & Parti (36), this conclusion is derived from the empirical results reported. For a review of the literature on peak-load demand and elasticities, see EPRI's "Time-of-Day" report (8).

substantially different short-run price elasticities and different appliance stock price elasticities. However, the calculated long-run price elasticity of electricity demand, and the short and long run income elasticities, are in close agreement because of compensating differences in the calculations.

Garbacz does not address the reasons for the differences in his results, and the only apparent explanation is the difference in price measures. One study (40) uses a measure of the average price of electricity to obtain a -0.19 short-run elasticity, while the other (41) uses a measure of the marginal price to obtain a -0.67 elasticity. The corresponding coefficients in the appliance stock equations also differ, but in a compensating fashion so that the long-run price elasticities are roughly equivalent. The explanation is questionable, however, in view of the evidence cited above that a distinction between average and marginal prices is not critical to the estimation of elasticities.

The Dubin & McFadden (42) model differs from that of Garbacz in two important respects. First, the appliance demand equation is structurally related to the utilization rate through the use of Equation 1, rather than as an additive component as in the Garbacz model. Second, Dubin & McFadden developed a specification of demand for appliance stocks based on the discrete nature of the purchase decision. Although popular in the literature on housing and transportation decisions,[17] this approach has had few applications to energy demand.

The Dubin-McFadden study is by far the most comprehensive model of energy demand in this survey. In the appliance choice model, the authors take into account the interactions among appliance choices (purchase of a washing machine, for instance, will increase the likelihood of purchasing a dryer), and between appliance choices and utilization rates (purchase of a dishwasher will increase the usc of a water hcater). However, the empirical part of this study is limited to the interactions between space-heating and water-heating decisions. The authors argue that the interactions require the use of consistent estimators, such as instrumental variables (IV) and TSLS.

Natural Gas

The issues involved in modeling residential demand for natural gas are essentially the same as those for electricity, except for the effect of natural gas price controls on the market. Both the controls and the market have been in a state of evolution. The effect of price controls shifted from relative ineffectiveness in the 1950s to increasing stringency, marked by curtailments, in the 1960s and 1970s. After the US Natural Gas Pricing Act of 1978, howcver, the price began to move toward market-clearing levels.

[17] For an excellent review of the literature on "qualitative response models," see Amemiya (43).

Over the same period, the number of localities served increased steadily, so that natural gas increased from a minor share of energy markets in the early 1950s to a major share in the 1970s. After mid-1970 the growth of the gas market was slowed by rising prices and supply curtailments.

The residential demand studies presented in Table 2 offer a variety of approaches to the issue of supply constraints. Blattenberger et al (33) make the most ambitious attempt to incorporate supply constraints directly into a demand model, with each US region weighted by a measure of natural gas availability. At the opposite extreme, Beierlein et al (31) and Hartman & Werth (35) ignore supply constraints. The middle ground is represented by Bloch (44) who restricts the scope of his sample to a community in New Jersey in which all houses have hookups.

REDUCED-FORM STATIC MODELS Bloch's study is the only reduced-form static model included, and the only one not mentioned in the previous section. Experimenting with the effect of three nonprice arguments (heating degree days, CPI, and time trend) on a pooled sample of household observations, he finds that the inclusion of either the time trend or CPI reduces the magnitude of the price elasticity to nearly one third.

Bloch's paper is of particular interest because of his unique sample of individual household observations over time. The model is not encumbered by a large number of control variables because of the near homogeneity of the housing stock and demographic characteristics across the households surveyed. Bloch finds that own-price responses vary substantially across households, contrary to what one would expect for homogeneous economic units, with disturbing implications for measuring demand behavior.

REDUCED-FORM DYNAMIC MODELS Beierlein et al (31) and Blattenberger et al (33) apply reduced-form dynamic models similar to their models of electricity consumption described above to residential natural gas consumption. The two studies differ in three primary respects: Beierlein et al use a sample of state observations restricted to the northeastern United States and an average price specification, while Blattenberger et al include all states, a marginal price specification, and a measurement of the supply curtailment. Beierlein et al obtain substantially higher price elasticities and lower income elasticities than do Blattenberger et al. The use of a curtailment variable by Blattenberger et al may account for part of the difference. The results of a postcurtailment sample with the correction for supply constraints are almost identical to a precurtailment sample, indicating that the curtailment variable performs well in correcting for supply constraints. In principle, at least, Blattenberger et al measure consumption responses due to changes in utilization rates and hookups, while Beierlein et al use a sample in which consumption adjusts only to changes in utilization rates.

Table 2 Residential natural gas

Study (date)	Sample[a]	Price elasticity[b] Short	Price elasticity[b] Long	Income elasticity Short	Income elasticity Long	Regression procedure[c]	Other variables[d]
Reduced-form, static							
Disagg., marg. price							
Bloch (1980)	1971–76; P; Twin Rivers, New Jersey (household)	−0.60				OLS	W (monthly observations)
		−0.22					W, time
		−0.24					W, CPI
		−0.22					W, CPI, time
Reduced-form, dynamic							
Agg., ave. price							
Beierlein et al (1981)	1967–77; P; N.E. states (state)	−0.23	−2.79	n.s.		OLS	PElec, POil
		−0.24	−3.17	n.s.		EC	
		−0.35	−3.44	n.s.		EC-SUR	
Agg., marg. price							
Blattenberger et al (1983)	1961–70; P; US (state)	−0.05	−0.33	0.11	0.77	VC	APNGas, PElec, POil, W
	1961–74	−0.032[e]	−0.26	0.06	0.48		APNGas, PElec, POil, W, gas supply
Reduced-form, end-use							
Agg., ave. price							
Hartman & Werth (1981)	1960–75; P; US (state)	n.s.		n.s.		WLS	ANGas
		n.s.		n.s.		FE	
		n.s.		n.s.		RE	

[a–d] See Table 1 notes.
[e] The figure is reported as −0.32, but is assumed to be a typographical error.

Two other findings are of interest. Beierlein et al test the importance of cross-correlations among residuals of separate equations for gas, electricity, and fuel oil. Using a combination of "error components" and "seemingly unrelated regression," they find that the results do not change substantially when compared to independent estimation. Blattenberger et al attempt to measure the relative importance of changes in the number of gas customers to changes in consumption per customer. They conclude that the price of gas is insignificant in explaining the number of gas customers, but very important in determining utilization rates per customer. While the first result is contrary to intuition, and should be viewed with skepticism, the implication holds that the composition of the capital stock (and hence relative appliance costs) are critical in determining initial fuel choice.

REDUCED-FORM END-USE MODELS Hartman & Werth (35) represents the only end-use model of residential natural gas available. Fitted to a pooled time series of aggregate state data for 1960–1975, the results are consistently insignificant for aggregate fuel consumption and generally insignificant for various end uses. Overall the estimates reveal no systematic relationship between gas prices and appliance stocks, and provide no real basis for choosing among the estimation procedures tested. The same poor performance was reported above with regard to residential electricity demand, for which it was noted that household-level data on appliance stocks makes a substantial difference in the performance of end-use models. Perhaps the most important conclusion that may be drawn from this study is that household-level data are required to estimate end-use models.

Fuel Oil

Data on consumption of fuel oil do not distinguish among consuming sectors, making it difficult to obtain reliable estimates of residential demand behavior (or that in any other sector). The only residential fuel-oil study considered here (see Table 3), by Blattenberger et al (33), estimates demand from state-level data.[18] The sample period is also of limited interest because it includes a single postembargo year. Depending upon model specification, Blattenberg et al obtain a short-run price elasticity of demand of −0.18 or −0.19. The effect of including and correcting for natural gas curtailments is slight.

COMMERCIAL AND INDUSTRIAL DEMANDS

The paucity of data on commercial and industrial energy consumption and related determinants limits the number and variety of studies of these

[18] Blattenberg et al do not present their data sources.

Table 3 Residential, fuel oil

Study (date)	Sample[a]	Price elasticity[b] Short	Price elasticity[b] Long	Income elasticity Short	Income elasticity Long	Regression procedure[c]	Other variables[d]
Reduced-form, dynamic							
Agg., marg. price							
Blattenberger et al (1983)	1964–74; P; US (state)	−0.19	−0.67[e]	0.12	0.44	VC	PElec, PGas, W, gas supply
		−0.18	−0.62	0.10	0.34 to 0.37		PElec, PGas, W

[a–d] See Table 1 notes.
[e] The figure reported in this study of −0.618 is inconsistent with the reported lag coefficient.

sectors. Data on energy consumption are generally available only for broad aggregates of commercial and industrial consumers. As a consequence, the possibilities for modeling demand behavior for individual or even moderately homogeneous units of consumers are limited. Widely diverse industries must be lumped together in ways that permit only crude inferences about the way energy is consumed and how consumption responds to changes in economic incentives.

Under these circumstances, it is not surprising to find that sophisticated models of commercial and industrial energy demand are not developed. Rather, most authors simply choose to modify the simpler, reduced-form models of residential consumption. Often, the only difference between residential and nonresidential models is in the substitution of output or value added for income as the measure of economic activity.

Electricity Demand

Tables 4 and 5 present the price and income elasticities from the six commercial and industrial electricity demand studies reviewed, separated into three model types: reduced-form static, reduced-form dynamic, and reduced-form end-use. The McRae & Webster (45) study is included (although it pertains to Canada) to permit comparisons with several similar models of US demand.

REDUCED-FORM STATIC MODELS Lyman (24), McRae & Webster (45), Kenney & Kershner (46), and Hirst et al (7) estimate similar reduced-form static models, but use very diverse samples. Hirst et al and Kenney & Kershner construct disaggregated samples of the commercial and industrial sectors respectively. Hirst et al use a cross-section sample to estimate the demand for electricity in institutional buildings in Minnesota; their sample of schools, hospitals, and local government buildings is a first step toward identifying subsections of the commercial market that would be expected to display greater homogeneity in energy use. Kenney & Kershner conduct their own survey of industrial establishments in New York State to obtain cross-section samples for three different years. Although not based on a random sample of establishments, the survey covers actual consumption behavior and management perceptions about such behavior. The authors also estimate an end-use model separately, which is described below.

Lyman (24) estimates both commercial and industrial demand from pre-1973 pooled time-series observations of selected utilities in the United States. His separate estimation of ten geographic regions can be compared to McRae & Webster's estimation of five Canadian regions. McRae & Webster (45) use a translog cost function model to estimate both a single

Table 4 Commercial electricity

Study (date)	Sample[a]	Price elasticity[b] Short	Price elasticity[b] Long	Income elasticity Short	Income elasticity Long	Regression procedure[c]	Other variables[d]
Reduced-form, static							
Agg., ave. price							
Lyman (1978)	1961–68; P; 10 regions (utility)	−0.27 to −4.56		not reported		MLE	PGas, D, W, G
Disagg., ave. price							
Hirst et al (1980)	1977–78; C; Minesota (building)		−1.05			—	PFuels, D, H (institutional buildings)
Reduced-Form, dynamic							
Agg., ave. price							
Uri (1978)	1971–75; T; US	n.s.[e]		n.s.		SUR	W, E, no. of customers (monthly observations)
Beierlein et al (1981)	1967–77; P; N.E. states (state)	n.s.		n.s.		OLS	PGas, POil
		n.s.		−0.15	−1.73	EC	
		n.s.		n.s.		EC-SUR	

[a–d] See Table 1 notes.
[e] The author regards the short-run estimate of −0.18 as significant (p. 239), but the standard error is reported as 0.96.

equation of pooled regional observations and separate time-series equations for each region. They also compare a pre-1973 sample with one that includes post-1973 observations. All four studies employ an average price measure.

In both the commercial and industrial sectors, Lyman (24) finds that price responses vary widely across regions (the same result noted above in regard to his residential electricity equation). The large variances preclude determination of whether commercial and industrial sectors differ significantly in terms of price responsiveness.

Despite similarities with Lyman's model, McRae & Webster (45) report significantly lower price elasticities and a smaller range of estimates for Canadian regions. Apart from sample differences, the results also appear to be sensitive to variations in model specifications. Lyman uses maximum likelihood estimation (MLE) to estimate separately the demand for electricity in each consuming sector, while McRae & Webster simultaneously estimate the demand for each fuel to account for correlations among end uses. McRae & Webster's results are sensitive to whether or not the data are pooled; they caution that the results from pooled data (reported in Table 5) do not reflect national estimates as well as do the regional time-series estimates. However, McRae & Webster obtain the counterintuitive result that the 1962–1976 sample produces *lower* elasticities than does the 1962–1973 sample, implying that industrial users became less responsive to higher prices. McRae & Webster speculate that the postembargo observations reflect incomplete adjustment to sharp price increases and may have lowered the apparent responsiveness to higher prices.

Kenney & Kershner (46) derive cross-sectional estimates that are consistently elastic, and less elastic in postembargo years. They find tremendous variance in electricity use patterns among firms, an indication of the difficulties inherent in those samples that aggregate across diverse industrial activities, even within limited geographic regions. Interestingly, their estimates of price responsiveness are larger than the survey respondents thought possible, given the technological constraints of production.

Hirst et al (7) estimate a demand elasticity for electricity for institutional buildings that is near unity. This result is probably a good indication of the demand for electricity for these uses, but its applicability to other commercial establishments such as warehouses and retail shops is questionable.

REDUCED-FORM DYNAMIC MODELS Both of the flow-adjustment models considered here use aggregate data, an average price variable, and a SUR

Table 5 Industrial electricity

Study (date)	Sample[a]	Price elasticity[b] Short	Price elasticity[b] Long	Income elasticity Short	Income elasticity Long	Regression procedure[c]	Other variables[d]
Reduced-form, static							
Agg., ave. price							
Lyman (1978)	1961–68; P; 10 regions (utility)	−0.27 to −3.52		not reported		MLE	PGas, D, W, G
McRae & Webster (1982)	1962–73; P; Canada (region)	−0.43				TSLS	PElec, PNGas, POil, PLPG, PCoal, PGas
	1962–76; P; Canada	0.00				2SLS	(All from single equation)
	1962–73; P; Quebec	−0.54					
	1962–76	−0.18					
	1962–73; P; Ontario	−0.65					
	1962–76	−0.18					
	1962–73; P; prairies	−0.60					
	1962–76	−0.18					
	1962–73; P; British Columbia	−0.55					
	1962–76	−0.18					
Disagg., ave. price							
Kenney & Kershner (1980)	1972; C; New York State (upstate) (plant)		−1.83		0.33	—	Income = manufacturing employment
	1975; C; New York State		−1.59		0.39		
	1978; C; New York State		−1.54		0.42		

Reduced-form, dynamic							
Agg., ave. price							
Uri (1978)	1971–75; T; US	−0.12	−0.12	0.87	0.87	SUR	W, E, no. of customers (monthly observations)
Beierlein et al (1981)	1967–77; P; N.E. states (state)	−0.10	[e]	n.s.	[e]	OLS	PGas, POil OLS
		−0.12	−3.55	n.s.		EC	
		−0.12	−2.97	0.04	1.00	EC-SUR	
Reduced-form, end-use							
Disagg., ave. price							
Kenney & Kershner (1980)	1972; C; New York State (upstate) (plant)		−1.09		1.04	—	For process heat
	1975; C; New York State		−1.34		0.95		
	1978; C; New York State		−1.02		0.89		
	1972; C; New York State (upstate) (plant)		−1.38		0.62		For driving motors
	1975; C; New York State		−1.44		0.66		
	1978; C; New York State		−1.77		0.62		
	1972; C; New York State (upstate) (plant)		−0.61		0.36		For lighting and space conditioning
	1975; C; New York State		−0.66		0.33		
	1978; C; New York State		−0.83		0.30		

[a–d] See Table 1 notes.

[e] Reported estimate of lag coefficient does not appear plausible and is inconsistent with text references.

procedure to estimate demand simultaneously among consuming sectors. It is not surprising, therefore, that they report essentially the same short-run price elasticity of -0.1 for the industrial sector. Both studies obtain statistically insignificant price elasticities for commercial demand. As was evidenced in the review of residential flow-adjustment models, the greatest disparity concerns the value of the adjustment coefficient: it is virtually zero for Uri (34) and nearly unity for Beierlein et al (31). Consequently, the long-run results reported are very different.

REDUCED-FORM END-USE MODELS Kenney & Kershner (46) are the only authors who disaggregate industrial electricity demand by end use, in a fashion similar to the residential appliance models discussed previously. Based on the same sample described above, they estimate price responses for three end-use categories—motor driving, heating, and lighting and space conditioning. The elasticities decline with the order of end uses listed, from -1.38 to -0.61 (for 1978 data). They find that total elasticities obtained from a weighted average of end-use elasticities are smaller than the results obtained directly from their reduced-form static equations. In part, this is to be expected because of the limited range of end uses considered.

Natural Gas

Market-clearing issues involved in modeling industrial and commercial demand for natural gas are similar to those noted in connection with the residential market, with two exceptions. First, supply curtailments are an even more significant factor since only nonresidential customers were actually disrupted. Second, many large firms negotiate individual supply contracts that may include curtailments during peak periods in exchange for lower prices. This gives the individual firm some control over price and complicates the identification problem.

The three natural gas models presented in Table 7 include a reduced-form static and a reduced form dynamic model of the familiar variety and a fuel-share model that estimates the proportional demand for natural gas. Only Beierlein et al (31) estimate both commercial and industrial demand (see Tables 6 and 7).

REDUCED-FORM STATIC MODELS McRae & Webster (45) apply their translog regional model of Canadian industrial energy demand to natural gas and obtain a pattern of results somewhat different from those reported for electricity. Overall, the estimated price elasticities of gas demand are much higher than for electricity demand and are generally above unity. The elasticities decline slightly when post-1973 data are introduced, but the decline is not as pronounced as in the case of electricity.

Table 6 Commercial natural gas

Study (date)	Sample[a]	Price elasticity[b]		Income elasticity		Regression procedure[c]	Other variables[d]
		Short	Long	Short	Long		
Reduced-form, dynamic							
Agg., ave. price							
Beierlein et al (1981)	1967–77; P; N.E. states (state)	n.s.		n.s.		OLS	PElec, POil
		−0.28	−1.86	n.s.		EC	
		−0.37	−2.27	n.s.		EC-SUR	

[a–d] See Table 1 notes.

Table 7 Industrial natural gas

Study (date)	Sample[a]	Price elasticity[b] Short	Price elasticity[b] Long	Income elasticity Short	Income elasticity Long	Regression procedure[c]	Other variables[d]
Reduced-form, static							
Agg., ave. price							
McRae & Webster (1982)	1962–73; P; Canada (region)	−1.30				TSLS	PElec, POil, PCoal, PLPG, PNGas, PGas
	1962–76	−0.71					
	1962–73; P; Quebec	−2.80					
	1962–76	−1.74					
	1962–73; P; Ontario	−1.56					
	1962–76	−1.15					
	1962–73; P; prairies	−1.44					
	1962–76	−1.02					
	1962–73; P; British Columbia	−1.75					
	1962–76	−1.33					
Reduced-form, dynamic							
Agg., ave. price							
Beierlein et al (1981)	1967–77; P; N.E. states (state)	−0.61	−2.40	0.78	3.08	OLS	POil, PElec
		−0.63	−2.51	0.75	2.95	EC	
		−0.62	−2.54	0.70	2.86	EC-SUR	
Fuel, share							
Agg., ave. price							
Chern & Just (1980)	1971, 1974; P; 19 states (state)		−2.16			GLS	POil, PCoal, manufac. costs ("marked share")
			−2.92				("total" or "conventional")

[a–d] See Table 1 notes.

REDUCED-FORM DYNAMIC MODELS Beierlein et al (31) extend their flow-adjustment model to commercial and industrial natural gas demand. As was the case in the residential sector, the results are apparently insensitive to the choice of estimation procedure. Commercial-sector price and income elasticities are considerably lower than their counterparts in the industrial sector, and a majority of the commercial estimates are statistically insignificant. Short-run industrial price and income elasticities are higher than those reported for industrial electricity use, a finding consistent with McRae & Webster's results for Canada.

FUEL-SHARE MODELS Energy consumption of both firms and households is generally comprised of more than one type of fuel. The demand for any given fuel may, therefore, be thought of as a function of both total energy use and the proportion of each fuel used. The model types discussed above can be modified to disaggregate the demand for a given fuel into total energy use and fuel share. In this way, a "fuel-share elasticity" can be derived that measures the change in a fuel's share of total energy use in response to a change in its price, with total energy use and the prices of other fuels held constant. Likewise, the elasticity of total energy use with regard to a change in the price of any given fuel may be calculated holding the fuel mix constant. For each fuel, the sum of "fuel share" of "total energy use" elasticities will equal the standard demand elasticity encountered in other studies.

Chern & Just (47) present estimates of a fuel-share model of natural gas demand for the pulp and paper industry in the United States, using a pooled sample of state-level data. The model postulates a two-stage production process whereby first the optimum fuel *mix* and then the optimum *level* of energy use is chosen. The conventional demand elasticity is decomposed into the effect of a fuel price change on the fuel share and the effect on total energy use. Fuel share is estimated simultaneously across fuel types. Chern & Just modify earlier versions of fuel-share models[19] by allowing the fuel share ratio of any two fuels to be affected by the price of a third fuel. The effect of the third price is found to be significant. The long-run fuel-share elasticity of 2.2 accounts for most of the total (or conventional) elasticity of −2.9, indicating that the pulp and paper industry responds to a change in the price of natural gas primarily by switching to alternative fuel sources and only secondarily by decreasing energy use (through lower output or energy conservation measures).

Fuel Oil

Estimation of the commercial and industrial demand for fuel oil, both distillate and residual, is hampered by the same data limitations observed in

[19] Baughman & Joskow (48) is the seminal model of this type.

the residential market. Each of the three models listed in Tables 8 and 9 use aggregate-level data, but are drawn from diverse sample populations.

REDUCED-FORM STATIC MODELS The regional fuel-oil demand estimates for Canada reported by McRae & Webster (45) exhibit a wider range of elasticities than either their electricity or natural gas results, from an inelastic -0.2 to a highly elastic -1.5. The larger regional variance may, in part, be accounted for by differences in fuel-oil availability and price levels among the regions.

REDUCED-FORM DYNAMIC MODELS The US Department of Energy (DOE) (49) reports the results of a flow-adjustment model for wholesale purchases of both residual and distillate oil by commercial and industrial users. Unfortunately, model and sample details are lacking. The long-run price elasticities of -0.5 and -0.7 are low compared to those of other industrial and commercial demand studies.

FUEL-SHARE MODELS Chern & Just (47) estimate a smaller price elasticity of demand for fuel oil than for natural gas by the pulp and paper industry, although both exceed unity. The fuel-share price elasticity is only slightly smaller than the total elasticity, as was the case with natural gas, indicating that industry response to a rise in the price of fuel oil primarily takes the form of fuel switching.

Fuel Consumption by Electric Utilities

Fuel consumption by electric utilities is commonly separated from that of other industrial uses because of dissimilar demand conditions and the presence of regulatory constraints on both price and fuel choice. The usual assumption of cost-minimizing behavior in the industrial sector is not appropriate for regulated utilities.

The only study included here—Uri (50)—is a translog fuel-share model. Although monthly data from 1972 to 1976 are used, only the 1976 own- and cross-price elasticities for each of ten DOE regions are reported. Uri acknowledges natural gas supply constraints and attempts to correct for the problem by adding a heating degree days (HDD) variable (because curtailments in some states and in some periods corresponded to the peak heating season). Uri's results are intended to measure the short-run responses in interfuel substitution (for dual-fired capacity) and intraplant substitution by changing the rank ordering of single-fired capacity. Uri finds that those regions with the greatest proportion of installed dual (or triple) capacity have the most elastic demand, while the lower elasticity estimates are found in regions where a single fuel represents a high proportion of total fuel costs. Although Uri concludes that utilities can be

Table 8 Commercial fuel oil

Study (date)	Sample[a]	Price elasticity[b]		Income elasticity		Regression procedure[c]	Other variables[d]
		Short	Long	Short	Long		
Reduced-form, dynamic							
Agg., ave. price							
EIA/DOE (1980)	1973–77; —; US		−0.7				P = wholesale (distillate)
			−0.7				(residual)

[a–d] See Table 1 notes.

Table 9 Industrial fuel oil

Study (date)	Sample[a]	Price elasticity[b] Short	Price elasticity[b] Long	Income elasticity Short	Income elasticity Long	Regression procedure[c]	Other variables[d]
Reduced-form, static							
Aggregated							
McRae & Webster (1982)	1962–73; P; Canada (region)	−0.23				TSLS	PElec; PNGas; PGas; PLPG; PCoal; POil
	1962–76	−0.36					
	1962–73; P; Quebec	−0.87					
	1962–76	−0.43					
	1962–73; P; Ontario	−1.07					
	1962–76	−0.43					
	1962–73; P; prairies	−1.51					
	1962–76	−0.32					
	1962–73; P; British Columbia	−0.92					
	1962–76	−0.85					
Reduced-form, dynamic							
Aggregated							
EIA/DOE (1980)	1973–77; —; US		−0.7				P = wholesale (distillate)
			−0.5				(residual)
Fuel-share							
Aggregated							
Chern & Just (1980)	1971, 1974; P; 19 states (state)		−1.28			GLS	PCoal, PNGas, wage, manufacturing costs (market share)
			−1.57				(Total)

[a–d] See Table 1 notes.

counted on to move in the "correct" direction in response to price signals, the fuel-share elasticities are all very small (with coefficients near -0.05), and not entirely consistent with the major shifts in fuel substitution observed in more recent years.

TRANSPORTATION

All of the studies in this section are concerned with the demand for gasoline. Gasoline sales data, moreover, do not permit a distinction among end uses or customer classes. Because some 70% of all gasoline is consumed by automobiles (51) and most of this is probably for noncommercial purposes, it is generally assumed that gasoline demand may be modeled according to characteristics associated with the residential sector. Hence, the models in this section are primarily concerned with noncommercial demand for gasoline. Related to this focus, a central issue in the models is whether or not the capital stock can be adequately represented by the stock of automobiles.

As with natural gas, gasoline supplics have been subject to price regulation and recurrent supply constraints that introduce identification problems. Legislated pollution controls also may be presumed to have shifted the supply curve, and possibly the demand curve, for automobiles. Some authors (though not included here) respond to these issues by restricting their samples to pre-1974 data, thereby limiting the usefulness of their results for assessing postembargo markets. In other studies, dummy shift variables have been used to incorporate changes in supply conditions. Only one author, Dahl (52), estimates a separate supply equation for gasoline.

Cross-substitution between automobiles and other forms of transportation, along with locational factors and demographic characteristics, are regarded by several authors as important to the demand for gasoline, but seldom appear in the models. The omissions are largely the result of inadequate data. At the same time, substantial regional variation in characteristics of gasoline demand is observed, but the source of that variation and the way to incorporate it into a model are unresolved issues.[20]

Reduced-Form Static Models

Two studies are reported in Table 10 that use reduced-form static models fitted to a pooled sample of aggregate state-level data. Aside from similarities in the samples, the studies differ in a number of respects.

[20] Principal papers in this debate are Greene (53), Mehta et al (54), and Kraft & Rodekohr (55).

Table 10 Gasoline

		Price elasticity[b]		Income elasticity			
Study (date)	Sample[a]	Short	Long	Short	Long	Regression procedure[c]	Other variables[d]
Reduced-form, static							
Agg., domestic							
Greene (1979)	1966–75; P; US (state)	–0.34		0.36		LSDV	G, auto, truck, D
Berzeg (1982)	1972–76; P; US (state)	–0.17		0.36		GEC	POP
Reduced-form, dynamic							
Agg., domestic							
Kwast (1980)	1963–77; P; US (state)	–0.07	–1.59	0.03	0.76	EC	G, E, POP
Berzeg (1982)	1972–76; P; US (state)	n.s.	n.s.	–0.18	–0.34	OLS	CPI, POP
		–0.15	0.14	0.42	0.40	GEC	CPI, POP
International							
Dahl (1982)	1970–78; P; 41 countries (country)	–0.13	–0.76	0.06	0.35	OLS	Auto
		–0.20	–1.00[e]	0.10[f]	0.50		Auto (low-price countries dropped)
Reduced-form, end-use							
Agg., static							
Reza & Spiro (1979)	1969–76; T; US (US)	–0.21	–0.33	0.60	1.44	GLS	Auto, D, E, MPG pollution, PAutos
Wheaton (1982)	1972; C; 25 countries		–0.74		1.26	OLS	Auto, MPG, T, D, G
Dahl (1979)	1936–41; 1947–72; 1975; T; US (US)	–0.44		not reported		2SLS	PAuto, auto, T, pollution, MPG

Disagg., static					
Archibald & Gillingham (1980)	1972–73; C; metro areas (household, residential)	−0.43	0.29	GLS	Auto, 1-car households
		−0.43	0.56		Auto, multi-car households
Archibald & Gillingham (1981)	1972–73; C; metro areas (household, residential)	−0.77	0.29	OLS	Auto, D, pollution, employ, 1-car households
		−0.22	0.56		Multi-car households
Agg., dynamic					
Paxson (1982)	1975–81; T; US (US)	−0.17	1.20	GLS	Auto, employ (1-year adjust)
		−0.07	0.91		Nonauto, employ (1 year)
		−0.14	0.56		Total (weighted average)

[a–d] See Table 1 notes.
[e] Value of −0.98 reported in text (p. 377) is not consistent with reported lag coefficient (p. 376).
[f] Value of 0.11 reported in text (p. 377) is not consistent with that reported in the text (p. 376).

Berzeg's model (56) is similar to Equation 4 while Greene's inclusion of a capital stock variable follows the modification reflected in Equation 4′. Greene (53) accounts for regional variation in factors such as climate and road conditions, aspects of demand that Berzeg (56) does not address, through the use of state dummy shift variables. In addition, Greene casts his model in per capita terms, while Berzeg is concerned with total sales by state. Both Berzeg and Greene are unable to obtain suitable results with OLS estimation. Greene uses a least-squares dummy variable technique and Berzeg uses a generalized error components (GEC) estimator. Unlike the EC model normally used to estimate models with pooled data, the generalized estimator does not constrain the error term to be independently and identically distributed. Despite the use of pooled observations, Greene interprets his results as short run. Nonetheless, Greene's income elasticity is identical to Berzeg's longer-run figure and his price elasticity is larger.

Reduced-Form Dynamic Models

Three flow-adjustment models of gasoline consumption are listed in Table 10, each fitted to pooled pre- and postembargo data. Kwast (57) and Berzeg (56) each use an EC procedure fitted to US state-level domestic observations while Dahl (58) uses OLS fitted to an international country-level survey. Dahl's study also differs in that it incorporates the number of automobiles and measures consumption on a per capita, rather than total, basis.

Berzeg is unable to obtain satisfactory results with either an OLS or GEC estimator. The OLS model produces an insignificant price coefficient. The GEC approach yields reasonable short-run price and income elasticities, but the lag coefficient is negative, giving the implausible conclusion that long-run responses to changes in price and income are smaller than the short-run responses. Berzeg consequently rejects the use of a flow-adjustment model in favor of the static model (presented above). Kwast (57) obtains very small estimates of both price and income gasoline elasticities (-0.07 and -0.03, respectively) from a model similar to Berzeg's. A large lag adjustment, moreover produces exceptionally large long-run estimates. Dahl's (58) price elasticity estimates are similar to those obtained from the reduced-form static models, although her income elasticities are somewhat lower.

Both Kwast and Dahl are interested in determining whether or not the price elasticity varies with the price level. Kwast tests for the occurrence of a structural shift in the US demand for gasoline as a result of the embargo-induced price shock. He is unable to find a significant difference in the price elasticity between pre- and postembargo observations. Dahl ranks the countries in her sample by price and tests for significant differences in the price elasticities among various groups of countries. The three countries with the lowest price exhibit a significantly lower price elasticity than do the other 38 countries. No other cross-country differences were discovered, nor were any significant differences found in income elasticities among countries. In comparing these results one should bear in mind that Kwast is testing for a structural change over time within a given sample population while Dahl is testing for structural differences among populations during a specific period of time.

Structural Models

The six structural models reviewed here are potentially more informative because they decompose the demand for gasoline into the demand for miles traveled, vehicle stock, and vehicle efficiency. To illustrate this relationship, note that the end-use equation (Equation 1) can be redefined for gasoline demand as

$$\text{Gas} = \frac{(\text{MILES})\cdot(\text{STOCK})}{\text{MPG}} \qquad 6.$$

in which miles traveled (MILES) is a measure of the utilization rate. The availability of miles per gallon (MPG) efficiency ratings makes it possible to quantify changes in the efficiency of the automobile stock (STOCK) and to estimate it separately.[21] It follows that

$$\ln \text{GAS} = \ln \text{MILES} + \ln \text{STOCK} - \ln \text{MPG}.$$

Hence, gasoline demand elasticities may be derived from the summation of the elasticities for these separate components.

Four studies use aggregate data to estimate gasoline demand. Reza & Spiro (59) and Paxson (51) use time-series samples of total US demand for gasoline while Dahl (52) uses a time-series of per capita US demand. All three studies include both pre- and postembargo observations. The fourth study, Wheaton (60), uses a pre-embargo international cross-section of country-level observations. Although this sample is not of particular interest here, the paper is included because the author compares the results of his structural model with those of a reduced-form static model fitted to the same sample.

Model specifications differ considerably among these four studies. In each case the stock of automobiles, automobile efficiency, and miles

[21] This measure is not possible in residential models, where information on appliance size and efficiency is unavailable. The effect of these qualitative changes on appliance stock characteristics is subsumed under the utilization rate measure.

traveled are endogenous, but the system of equations by which their interrelationships are defined differs. For example, Dahl incorporates the automobile stock variable in the demand for gasoline equation directly, while in Wheaton's model the stock variable operates indirectly through its effect on the demand for miles traveled. In addition to the endogenous demand-side variables, Dahl (52) incorporates a separate supply equation for gasoline. Both Paxson (51) and Reza & Spiro (59) incorporate measures of the demand for new automobiles and the depreciation rate as a means of estimating changes in the automobile stock. Paxson also introduces a lagged quantity variable to create a dynamic structural model. Because observations are monthly, she interprets the lagged adjustment "long-run" results as annual. As for estimators, Dahl uses two stage least squares, while Paxson and Reza & Spiro use generalized least squares to eliminate serial correlation among the error terms. Wheaton (60) does not find any correlation among the error terms and, therefore, chooses OLS estimation.

Paxson (51) obtains the lowest short-run (one year) price elasticity (-0.17) of these structural models. When the "stock" variable is redefined to include trucks as well as automobiles, the estimate is even lower. Dahl's short-run price elasticity (-0.44) is twice that of Reza & Spiro, despite model similarities. Dahl's somewhat larger estimate may be due in part to inclusion of observations dating back to 1937. Wheaton reports only long-run results from his international sample, including a price elasticity that is about the same as that estimated by Dahl (58) from a similar sample using a reduced-form model (see above). Both Wheaton and Reza & Spiro obtain highly elastic long-run income responses. Unlike Paxson, Wheaton finds that redefinition of the stock variable to include trucks as well as automobiles does not significantly alter the results.

In two separate studies, Archibald & Gillingham (61, 62) present the only available estimates of gasoline demand using disaggregated data. Employing the 1972–1973 Consumer Expenditure Survey (63) in both studies, they divide their cross-section sample into single- and multi-car households, on the premise that multi-car households might exhibit greater flexibility in responding to price changes because they can adjust the relative use of each vehicle. They distinguish between the demands for miles traveled, efficiency, and the automobile stock, but do not estimate this last variable.

Archibald & Gillingham interpret the results of each study as short run, in which case the price elasticities obtained are higher than those of the other structural models reviewed above, while income elasticities are about the same. Because the authors do not appear to have accounted for stock variation among households (only within households), these results might better be viewed as longer run. In the 1980 study (61), the demand for gasoline is calculated directly by GLS. The price elasticity of demand is

-0.43 for each group when estimated at the sample means (divergences occur away from the means) while the income elasticity is nearly twice as high for multi-car households. Interpreting income elasticities is complicated because the multi-car group likely has a higher average income, if not a wider variance in income. The 1981 study (62) differs in several respects. Using OLS, and including employment and variables measuring the effect of pollution regulations on supply, the study obtains dramatically changed results. Single-car households now exhibit a price elasticity three times higher than that of multi-car households. This disparity is surprising and contrary to the initial assumption that multi-car households should exhibit a more flexible price response. Strangely, the authors do not comment on this disparity.

The structural models reviewed here are designed primarily to identify the component sources of demand changes. Table 11 reveals very little agreement among the studies as to the relative importance of each factor. The short-run demand elasticity for travel with respect to the price of gasoline, for instance, ranges from -0.08 to -0.61. The various authors may or may not have broken out efficiency as a short-run response; where it is not considered [Reza & Spiro (59) and Paxson (51)], the implicit assumption is that consumers cannot affect miles-per-gallon in the short run. In each study, the price of gasoline has a greater effect on demand through miles traveled than through either of the other variables in both the short and long run. Wheaton (60) finds that in the long run, the effect of a change in income operates principally through changes in the automobile stock, as might be expected. Reza & Spiro (59), on the other hand, conclude that the income effect operates principally through the number of miles traveled per vehicle (MILES). This result is due, at least in part, to the nature of their model; Reza & Spiro do not break out MPG as a separate variable, so its effect on gasoline consumption is contained in the MILES variable, increasing the significance of miles traveled in elasticity estimates. Finally, the results indicate that efficiency, unlike the other components, is an inferior good.

The entry for Dahl (52) in Table 11 illustrates another estimation issue. When the separate estimates of the contribution of travel, stock, and MPG to gasoline consumption are combined, they do not equal total consumption. This difficulty was encountered by Kenney & Kershner (46) in their industrial electricity study discussed above and should be expected, given sampling errors and possible misspecifications of the component equations. To insure consistency, Equation 6 may be used to calculate the third component of demand. Most of the other authors estimate only two of the three component demands independently, thereby ensuring consistency without addressing the issue raised by Dahl of whether or not the separate effects would add to total consumption.

Table 11 Decomposition of gasoline demand elasticities

	Short-run gas price elast.			Long-run price elast.			Short-run income elast.			Long-run income elast.		
Study (date)	MILES	STOCK	MPG	MILES	STOCK	MPG	MILES	STOCK	MPG	MILES	STOCK	MPG
Reza & Spiro (1979)	−0.21			−0.20	−0.13		0.60			0.83[a]	0.61	
Wheaton (1982)												
25 countries, nominal				−0.50	n.s.	0.32				0.54	1.38	−0.21
25 countries, deflated				−0.54	n.s.	0.33				0.46	1.89	−0.20
42 countries, nominal				−0.55	n.s.	0.26				0.33	1.43	−0.12
Archibald & Gillingham (1981)												
One car	−0.61		0.16				0.23		−0.06			
Multi-car	−0.16		0.06				0.47		−0.08			
Paxson (1982)												
One year	−0.14						0.56					
Dahl (1979)												
MILES econometrically estimated												
(1936–72 data)	−0.08		0.21									
(1936–74 data)	−0.2[b]		0.08									
MILES not econometrically estimated												
(1936–72 data)	−0.23		0.21									

[a] The figure of 1.44 reported in Reza & Spiro (59), p. 312, is for total miles traveled: this figure is adjusted for the change in the capital stock to obtain a miles-per-automobile figure and for consistency with the other studies.
[b] Assumed from statement in text, p. 430.

COMPARISONS AMONG THE RESULTS

This section looks at general characteristics of the literature on energy demand behavior rather than at individual studies. Empirical results of the studies reviewed above are compared among end-use sectors, fuel types, and modeling issues, and with the results of earlier studies reviewed by Bohi (3). The comparative judgments made here are highly subjective, but are based on a number of specific criteria. In addition to the usual measures of statistical significance and conformity among studies, results from studies that use highly disaggregated data, that model the structural elements of demand behavior, and that are generally consistent with the economic theory of demand are considered more reliable.

Elasticities Across Consuming Sectors and Fuel Types

Table 12 presents our interpretation of the consensus estimates of price elasticities by fuel type and consuming sector, based on the studies reviewed above. In several cases there are not enough independent estimates on which to base a conclusion, or the range of estimates is so wide that the elasticity must be considered uncertain. The consensus estimates should be viewed with caution, of course.

Perhaps the most disconcerting conclusion to be drawn from Table 12 is the large amount of uncertain results. Outside of residential demand for electricity, the quality of the information obtained from years of study ranges from weak to very poor. Information on commercial and industrial energy demand is weakest, and research conducted in recent years adds little to that reported earlier (3).

Residential demand for electricity appears to have a price elasticity near -0.2 in the short run and near -0.7 in the long run. The long-run estimates vary from one study to the next, but overall they indicate that the elasticity is less than unity. Because the consensus estimate is similar across a wide variety of models and samples, our conclusion for residential electricity is considered more reliable than that for other fuels and sectors. Residential

Table 12 Consensus estimates of price elasticities (in absolute value)

	Electricity Short/long	Natural gas Short/long	Fuel oil Short/long	Gasoline Short/long
Residential	0.2/0.7	0.2/0.3	uncertain	
Commercial	uncertain	uncertain	uncertain	
Industrial	uncertain	uncertain	uncertain	
Transportation				0.2/< 1.00

natural gas demand appears to respond to price to the same extent as electricity in the short run, but is substantially less elastic in the long run. These results are not surprising since there are similar constraints on variations in utilization rates and a wider range of substitution possibilities for electricity (natural gas is restricted primarily to space and water heating). In addition, fuel switching in favor of natural gas in residential markets has been limited by institutional constraints on hookups during the 1970s.

Very little information is conveyed about commercial and industrial energy demand behavior. Based on available evidence, both commercial and industrial demand seem to be more elastic than residential demand across all fuels. However, the evidence is very tenuous. Until observations at the firm level become available, there is little hope for a better understanding of demand behavior in these sectors.

The demand for gasoline appears to have a short-run price elasticity on the same order as that for electricity and natural gas, and a long-run elasticity slightly less than unity. Substantial variation in reported estimates precludes a more precise conclusion about the long-run figure. The income elasticities reported for gasoline (not shown in Table 12) are generally larger than those for other fuels and sectors. Gasoline consumption appears to be a normal good, or perhaps even a superior good, in contrast to other fuels.

As a final observation, we note an apparent inverse relationship between price and income elasticities across the different models: Models that give relatively large price coefficients also tend to give relatively low income coefficients, and vice versa. The inverse relationship may result from an interdependence between price and income effects enforced by linear relationships, as suggested by Deaton (64), or by the nature of the demand process because of indirect income and price effects.

Pre- and Postembargo Demand

The dramatic increase in energy prices in 1974, following years of stable or declining real prices before 1974, provides an opportunity to test a number of hypotheses about demand behavior. One such hypothesis is that demand becomes more elastic at higher price levels. Another is that rapid price changes produce structural shifts in demand related to nonprice variables or to nonquantifiable influences such as a conservation ethic.

To determine if a change in energy price elasticities has occurred, we compare the consensus estimates in Table 11, which are based on post-1974 observations, with those reported to Bohi (3), which use samples generally limited to the pre-1974 period. The latter estimates are summarized in Table 13. Comparing Tables 11 and 12, it is apparent that there is no overall trend toward larger price elasticities in the more recent group of studies.

Variances exist in the reported estimates for individual fuel uses, but the differences are slight and may be attributed to sampling errors.

Several authors have tested for structural changes within their own samples. One procedure is to partition the sample and to compare the estimates of different subsamples. Using this procedure to test for structural change in the gasoline market, Kwast (57) is unable to reject the null hypothesis of no structural change resulting from the 1973 price shock and Paxson (51) is unable to reject the same hypothesis for the 1979 price shock. In contrast, McRae & Webster (45) compare estimates of industrial demand for different fuels fitted to samples covering 1962–1976, and find that the addition of postembargo observations produces less-elastic results for each fuel type, contrary to expectations. Kenney & Kershner (46) also find a drop in the price elasticity of industrial demand for electricity in their 1978 sample compared to their 1972 sample. Blattenberger et al (33) divide their 1960–1975 sample of residential electricity consumption into periods of rising and falling prices and into periods of slow and rapid price escalation. They are unable to identify a significant difference among the subsamples on either basis.

Another approach used in several studies incorporates the effect of an event such as the oil embargo directly into the model with dummy variables. An additive dummy is used to reflect a shift in the intercept term, while a multiplicative dummy variable is used to measure the shift in the slope coefficient of a variable included in the equation. In the gasoline market, Reza & Spiro (59), Kwast (57), and Greene (53) find that an additive dummy variable proves significant for 1974, indicating that consumption is adjusted downward for the effect of the embargo. Among the residential electricity models, Blattenberger et al (33) and Uri (34) find that an additive dummy proves significant for 1974. None of the tests employing a multiplicative dummy variable indicated a structural shift.

An interesting picture tends to emerge from the separate tests for changes in demand induced by the oil crisis. In general, the additive dummy variables indicate a significant shift in demand, while the variable

Table 13 Consensus estimates of price elasticities, Bohi study (3) (in absolute value)

	Electricity Short/long	Natural gas Short/long	Fuel oil Short/long	Gasoline Short/long
Residential	0.2/0.7	0.1/< 0.5	uncertain	
Commercial	uncertain	uncertain	uncertain	
Industrial	uncertain	uncertain	uncertain	
Transportation				0.2/0.7

coefficients indicate no significant changes in elasticities. The divergence in results may be regarded as somewhat inconsistent, since elasticities change with base values when there is a shift in the entire schedule. Nevertheless, as a tentative conclusion, the energy crisis does not appear to have affected the structural characteristics of demand as related to the usual economic determinants; rather the changes have occurred in those determinants of demand that are incorporated in constant terms and error terms. This conclusion is tentative because of the relatively short period that has elapsed since 1974. The hypotheses deserve reexamination as more data become available.

Implications for Modeling Issues

Several characteristics have been used to distinguish among the models reviewed above, though for some (e.g. functional form and estimation technique) there is insufficient information to make comparisons among alternative approaches. This section summarizes our observations concerning model structure, level of aggregation, measurement issues, and supply considerations.

MODEL STRUCTURE Less can be said about the importance of modeling approaches than might be expected because other factors cloud the comparisons. As a group, the structural models perform better than the reduced form models; they provide more information, produce statistically significant results, and exhibit greater consistency across the separate studies. However, the structural models are almost always fitted to highly disaggregated data. The subgroup of static reduced-form models fitted to disaggregated data (in the residential electricity sector) also perform well and yield elasticities comparable to those of their structural counterparts.

Among the reduced-form models, dynamic models produce highly erratic elasticities because of variations in the lag-adjustment coefficient. Also for this class of models, intuitively important variables are more frequently registered as statistically insignificant. The reduced-form end-use models, used principally in the residential electricity sector, yield implausibly high short-run price elasticity estimates. The reduced-form static models fitted to highly aggregated data yield relatively high long-run elasticities.

LEVEL OF AGGREGATION Studies employing highly aggregated units of observation are plagued with a large number of insignificant price and income coefficients and, where comparisons are possible, produce larger price elasticities than do disaggregated samples. Because the comparisons are limited for the most part to residential electricity demand, any conclusions associated with differences in samples must be tentative. Nevertheless, from the evidence so far, the level of aggregation of the data appears to be the single most important factor that accounts for variations in results among the studies.

MEASUREMENT ISSUES Two broad categories of explanatory variables are used in the models surveyed: those derived from the economic theory of demand, which are applicable to every sector and fuel type (e.g. prices and income), and those that are intuitively important determinants of demand for a particular fuel and sector. The latter variables are often chosen on an adhoc basis and are reported in final results if they prove to be statistically significant. Demographic characteristics and building characteristics are significant in residential models, but generally show up only when fitted to disaggregated data. Weather variables are frequently employed in both residential and commercial models, although they perform best in the residential sector. Gasoline demand models often include variables measuring geographic characteristics such as land area and miles of highway. Despite their intuitive appeal, however, these variables do not account for regional variations in demand behavior.

Among the variables suggested by economic theory, price naturally receives the most attention. All of the studies consistently agree that energy demands are systematically determined by relative prices, though they disagree on the degree of influence. For electricity and natural gas, attention is focused on the appropriate measure of the own-price elasticity. With few exceptions, the studies show no appreciable differences between price responses associated with average or marginal prices. Other parameters of declining rate schedules remain in question, however, such as fixed charges and inframarginal rates.

The choice of a measure of economic activity depends on the consuming sector under consideration. In the residential sector, either personal income or expenditure is used as the budget constraint. Though the latter measure is probably closer to a measure of permanent income, a comparison of parameters among similar models does not reveal any substantial differences in results related to the measure of income. In the commercial sector, retail sales are generally used as the measure of economic activity, while value added is used in the industrial sector. In all cases, the role of the activity variable is questionable. Activity variables perform best with respect to the residential sector, and insignificant results are limited to models fitted to aggregate data.

Most studies also include the price of at least one competitive fuel among the independent variables. Frequently, cross-price effects are found to be insignificant, and where significant, are typically small. These poor results are perhaps more a reflection of the partial equilibrium class of models under examination than the absence of true cross-price effects.

SUPPLY CONSIDERATIONS Few authors take supply conditions into account in estimating demand behavior. Those who treat price as simultaneously determined by supply and demand conditions indicate that the issue deserves consideration. Supply conditions are usually incorporated into a model when specific constraints (such as those resulting from interruptible service contracts or product price regulations) are assumed to be operative. As such, the method of modeling supply varies with the market and time period under consideration. There is, however, no clear evidence of a systematic simultaneous equations bias in estimates of price elasticities.

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